SUSHIL CHAVAN

AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with project experience spanning **Machine Learning**, **Deep Learning**, **Natural Language Processing (NLP)**, and **Generative AI**, including LLM-powered analytics, Retrieval-Augmented Generation (RAG), semantic search, and computer vision. Internship experience delivering **Power BI** dashboards and **SQL/Python** data pipelines. Comfortable across the applied AI stack: data preprocessing, feature engineering, model evaluation, and Flask-based backend integration for deployment-ready systems. B.Tech graduate (Artificial Intelligence & Machine Learning) looking to bring this foundation to an AI/ML Engineering or GenAI Engineering role.

KEY SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Data Cleaning, Feature Engineering, EDA, Statistical Analysis

Machine Learning / Deep Learning: Scikit-learn, TensorFlow, Keras, CNN, ANN, Regression, Classification, Clustering, Model Evaluation, Model Optimization, OpenCV

Generative AI / NLP: LLMs, LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Transformers, Sentence Transformers, Embeddings, Semantic Search, Vector Search

Backend & Deployment: Flask, Django, Streamlit

Databases: MySQL, FAISS (Vector Database)

Visualization & BI: Power BI, DAX, Advanced Excel (Pivot Tables, VLOOKUP, XLOOKUP), Matplotlib, Seaborn

Tools: Git, GitHub, Jupyter Notebook, VS Code

EXPERIENCE

Data Science Intern — Gamaka Intellectual Machines Nov 2025 – May 2026

Pune, India

- Engineered interactive **Power BI** dashboards using **DAX** and data modeling to support business reporting and KPI tracking for stakeholders.
- Built data cleaning and analysis workflows in **Python, SQL, Pandas, and NumPy**, improving data quality and surfacing actionable business insights.
- Performed **exploratory data analysis (EDA)**, statistical analysis, and Matplotlib-based visualization to identify trends and support data-driven decision-making.
- Applied **Scikit-learn** and **MySQL** in collaborative machine learning and database workflows, using **Git** for version control.

PROJECTS

LexiMind — Legal Document Intelligence System [GitHub]

Technologies: *Python, Scikit-learn, Sentence Transformers, Flask, Pandas, NumPy, Cosine Similarity, NLP, Semantic Search*

- **Problem:** Legal teams needed a faster way to classify and retrieve relevant judgments from a large, unstructured PDF archive.
- **Solution:** Led the ML, NLP, and backend integration modules (5-member team) to build an automated pipeline that extracts and structures metadata — case titles, citations, bench details, legal categories — from **17K+ legal judgment PDFs**.
- **Modeling:** Trained a **TF-IDF + Logistic Regression** classifier for multi-class legal judgment categorization, reaching **~92% accuracy**; generated semantic embeddings with **Sentence Transformers (all-MiniLM-L6-v2)** and used cosine similarity for Top-5 relevant-judgment retrieval.
- **Deployment:** Integrated trained models into a **Flask** backend supporting PDF upload, text extraction, category prediction, and semantic search, with inference and memory usage optimized via **Joblib** for a modular, deployment-ready architecture.

Data Analyst AI Agent [GitHub]

Technologies: *Python, Flask, Gemini API, Pandas, NumPy, Matplotlib, LLMs, EDA*

- **Problem:** Non-technical users need a way to explore datasets and get insights without writing code.
- **Solution:** Built an AI-powered data analytics assistant that converts natural-language queries into executable **Pandas** code using the **Gemini API**, executes it server-side, and returns results through a **Flask** web interface.
- **Automation:** Automated core EDA steps — null-value handling, duplicate removal, descriptive statistics, data-type inspection, and dataset summarization — plus Matplotlib-based visual reporting.
- **Impact:** Enabled conversational, code-free dataset exploration for non-technical business users.

Face Mask Detection System [GitHub]

Technologies: *Python, TensorFlow, Keras, CNN, OpenCV, Streamlit, NumPy*

- **Problem:** Real-time compliance monitoring for mask usage from image input.
- **Solution:** Developed a **CNN**-based deep learning model achieving **98–99% validation accuracy** for multi-class mask detection (Mask Worn / No Mask / Incorrect Mask).
- **Pipeline:** Built an image preprocessing pipeline (face detection via **OpenCV Haar Cascade**, cropping, resizing, normalization, data augmentation) to improve generalization.
- **Deployment:** Shipped a **Streamlit** web app for real-time image upload and inference with confidence scores.

Bank Term Deposit Subscription Prediction GitHub

Technologies: *Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Logistic Regression, KNN, Naïve Bayes, Random Forest*

- **Problem:** Predict which bank customers are likely to subscribe to a term deposit, using **41K+ banking records**.
- **Approach:** Ran EDA, categorical feature analysis, label encoding, and feature scaling; addressed class imbalance with class-weighted learning.
- **Evaluation:** Benchmarked **Logistic Regression, KNN, Gaussian Naïve Bayes, and Random Forest** using Accuracy, Precision, Recall, F1-score, and Confusion Matrix to select the best-performing classifier.
- **Insight:** Analyzed Random Forest feature importances to surface the customer attributes most predictive of subscription, informing marketing targeting.

HACKATHON

Internal Smart India Hackathon (SIH 2025) — AI-Based Product Compliance Verification System

Technologies: *Python, OpenCV, OCR, Web Scraping, BeautifulSoup, Requests, Pandas, Flask*

- Built an AI-powered compliance verification system that extracts product information from images (**OCR**) and product web pages (**web scraping**).
- Automated validation of mandatory product information against predefined compliance requirements to speed up and improve accuracy of compliance checks.
- Structured extracted product data to flag missing or incomplete information and generate compliance reports.

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning May 2026

G H Raisoni College of Engineering and Management, Jalgaon

CGPA: 6.9

Higher Secondary Certificate (HSC), Science — Shri Shivaji Vidyalaya, Buldhana Mar 2022

79.83%

Secondary School Certificate (SSC) — Bharat Vidyalaya, Buldhana Mar 2020

76.40%

CERTIFICATIONS

- Gen AI 101 — IT-ITeS SSC NASSCOM
- Gen AI Tools — IT-ITeS SSC NASSCOM
- Python 3 for Absolute Beginners: From Scratch to Advanced Level — Udemy
- Introduction to Artificial Intelligence — Infosys Springboard